APPLICATION NOTE

Reading an inspection report

1 Scope

For purchasing and quality departments receiving a serial-numbered target. This note explains how to read the inspection report for sample unit 000311, choose the appropriate supporting record, and retain the delivery evidence. It is operational guidance, not legal advice.

2 Report, CoC and certificate compared

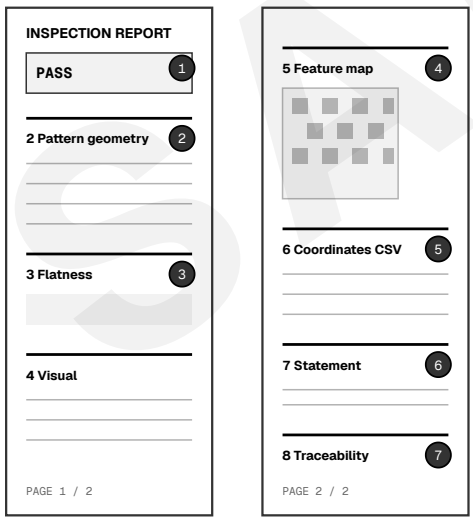
DOCUMENT	ISSUER	ACCREDITATION	WHAT IT DEMONSTRATES	WHEN YOU NEED IT
CT-IR Inspection report	Calib.Tools	No — manufacturer record	A specified board was inspected against its datasheet at the time of inspection, with measured data.	You need serial-level receiving evidence or engineering data.
CT-CoC Certificate of conformance	Calib.Tools	No — manufacturer statement	Listed items were manufactured to the referenced drawing, inspected per procedure, and conform to the datasheet.	A purchase order or customer asks for a conformance statement.
ILAC-MRA certificate CT-CC cover	Accredited third-party laboratory	Yes — laboratory scope applies	The laboratory's calibration result and stated uncertainty for the item or service in scope.	Your quality plan, customer, or regulator requires an accredited laboratory certificate.

DECISION POINT

Choose the record named by your purchase order or quality plan. A CT-IR gives serial-level inspection evidence; a CT-CoC gives a manufacturer's conformance statement; an ILAC-MRA certificate comes from an accredited third-party laboratory.

3 The report section by section

CT-IR-000311



Miniature schematic. Callouts align with the reading guide.

CALLOUT	WHAT TO READ	CT-IR-000311 EXAMPLE
1	Overall result sets the decision for the report's defined checks.	PASS — all measured features within datasheet tolerance.
2	Compare deviation with tolerance; geometry values are in µm unless marked otherwise.	Pitch X +1.2 µm vs ±2; max single-feature deviation 1.6 µm vs ≤ 2.
3	Confirm measured flatness, method, point count, and limit.	12 µm, 25 points, limit ≤ 20 µm.
4	Feature map establishes the coordinate frame and locates the largest vector.	X right, Y down; corner id 88; 1.6 µm.
5	Use the complete CSV for analysis, not only the ten-row preview.	176 rows in CT-CSV-000311.csv.
6	Read the statement's scope and date before reusing it in a supplier file.	References CT-DS-CHA-C Rev A at time of inspection.
7	Keep the instrument and reference record with the report.	OCMM-01; raw data retained for 10 years.

The report is evidence of inspection. It does not substitute for an accredited laboratory certificate where your process requires one.

SAMPLE

4 Incoming inspection checklist

Record these checks in the receiving process; leave acceptance criteria and approvals to your own quality system.

RECEIVED TARGET

- ☐ Match SN 000311 to the target.
- ☐ Match the part number to the PO.

REPORT RECORD

- ☐ Check CT-IR-000311 · Rev A.
- ☐ Confirm PASS belongs to that serial.

DATA PACKAGE

- ☐ Save CT-CSV-000311.csv (176 rows).
- ☐ Check X right / Y down frame.

RELEASE RECORD

- ☐ Record checksum and verify URL.
- ☐ Escalate missing files to quality owner.

5 Archiving for ISO 9001

Use the report and CSV as evidence supporting control of monitoring and measuring resources under ISO 9001:2015 § 7.1.5. Retain them with the asset or receiving record.

Required set	`CT-IR-000311.pdf` + `CT-CSV-000311.csv`
Online record	`calib.tools/verify/000311`
Supplier raw data	Calib.Tools retains raw data for 10 years under the serial number.
Your QMS	Set retention, acceptance, and access controls with the quality-system owner.

QUALITY-SYSTEM BOUNDARY

This application note does not provide legal advice. Ask your quality-system owner how ISO 9001:2015 applies to your process.

6 Checking file integrity

Verify the extracted package against its SHA-256 manifest. To review the coordinate data, recompute the largest radial deviation from columns `id, row, col, x_nom, y_nom, x_meas, y_meas, dx_um, dy_um`.

SHELL

```
# Run from the extracted delivery folder
sha256sum -c CT-FM-1047-01.sha256
```

PYTHON / PANDAS

```
import pandas as pd
df = pd.read_csv("CT-CSV-000311.csv", comment="#")
df["dev_um"] = (df.dx_um**2 + df.dy_um**2) ** 0.5
print(df.loc[df.dev_um.idxmax(), ["id", "dev_um"]])
```

7 References

REFERENCE	USE
calib.tools/verify	Retrieve report and companion files by serial number.
CT-AN-001	Pattern files and software setup.
CT-UM-GEN	General handling and use.
ISO 9001:2015	Clause 7.1.5 — monitoring and measuring resources.
ILAC MRA	Mutual recognition context for accredited laboratory certificates.

NEED AN ACCREDITED LABORATORY CERTIFICATE?

Where required, request the applicable ILAC-MRA option for the serial-numbered target and retain the original laboratory certificate with your quality record.

calib.tools/files/CT-AN-004

SAMPLE